

Title

Community health worker-mediated remote clinical care for hypertension in rural African settings: design of a mixed-methods implementation trial

Abstract

Background: Hypertension is increasingly prevalent in sub-Saharan Africa, yet rural populations face substantial barriers to clinic-based care including distance, transportation costs, and limited facility capacity. Community health worker (CHW)-coordinated remote consultation with clinicians represents a potential strategy to extend care access while maintaining quality. **Methods:** The SEARCH Sapphire study is a mixed-methods implementation trial in rural western Kenya and southwestern Uganda. Adults aged ≥ 40 years with moderate-severe hypertension will be randomized to CHW-facilitated remote clinical consultation (intervention) or usual clinic-based care (control). The trial incorporates both quantitative measurement of blood pressure control and health system outcomes, and qualitative exploration of acceptability through in-depth interviews with patients ($n \sim 40$) and healthcare providers ($n \sim 15$). Qualitative data will examine perceived quality, accessibility, integration into existing workflows, and barriers/facilitators to implementation. Mixed-methods integration will assess whether CHW-remote models are effective, acceptable, and sustainable for rural hypertension management.